

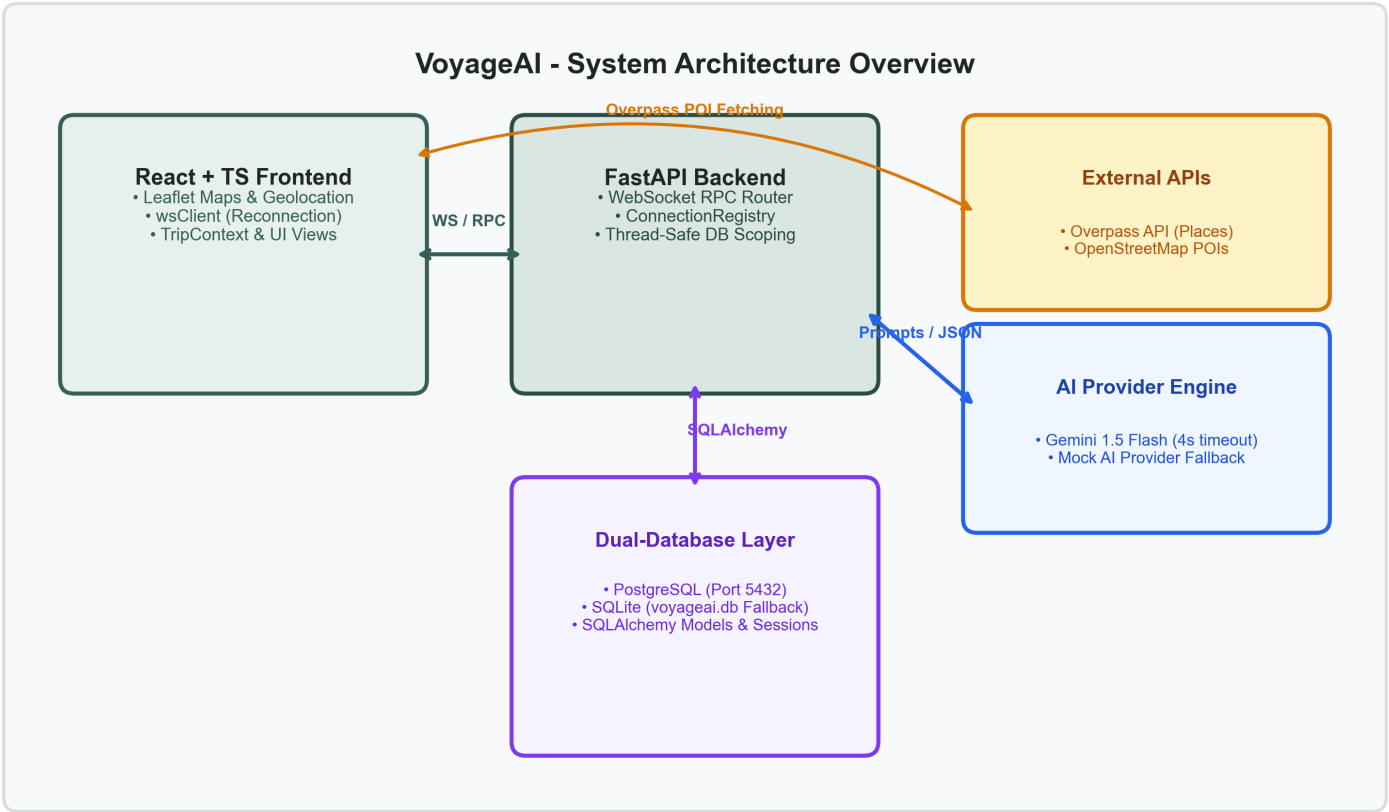
VoyageAI Architecture & Skills Report

A Simplified Technical Guide to System Architecture, Data Flows, and Skills Used

Executive Summary: VoyageAI is a full-stack, real-time travel planning platform. It leverages AI-driven itinerary generation (Google Gemini 1.5 Flash with Mock AI fallback), real-time location-aware POI discovery (OpenStreetMap Overpass API), dual-database failover (PostgreSQL + SQLite), and low-latency WebSockets RPC with exponential backoff reconnect and offline request buffering.

1. High-Level System Architecture

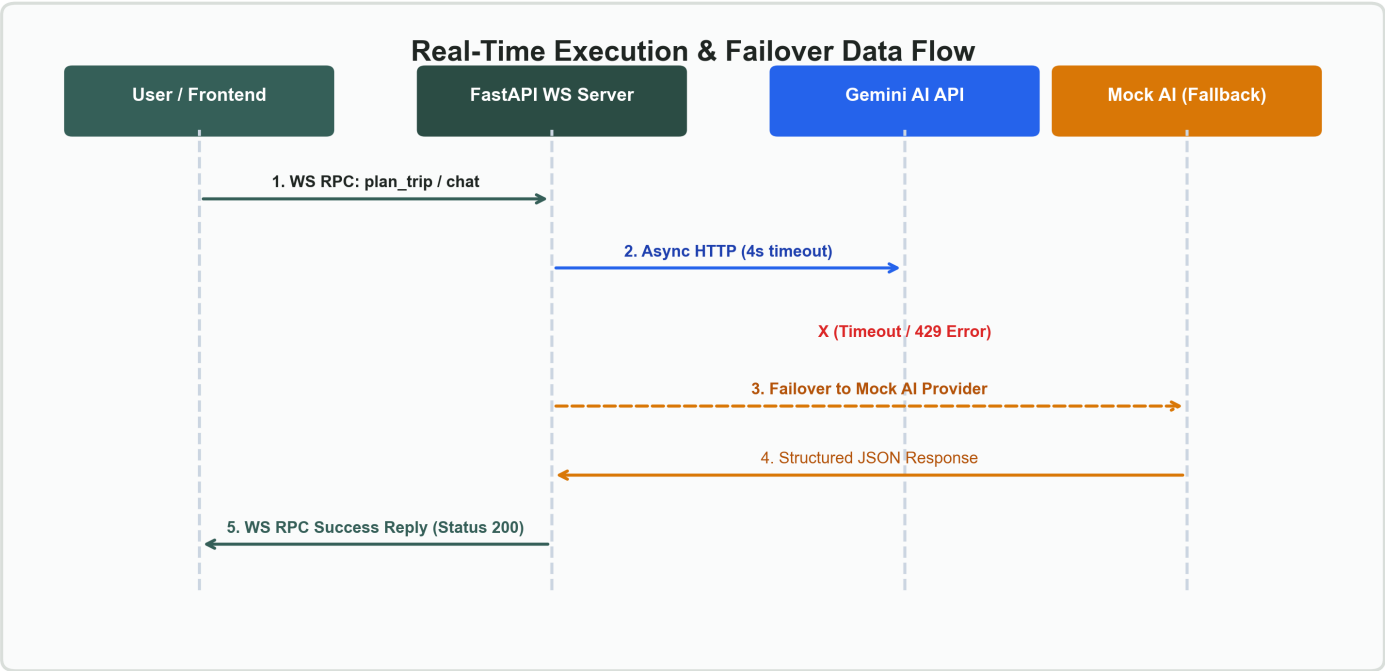
The VoyageAI ecosystem is split into 5 core layers designed for high responsiveness and fault tolerance:



System Layer	Component Tech	Role & Architecture Responsibilities
Frontend Client	React 18 + TS + Vite Leaflet Maps	Renders SPA, user location POI discovery, interactive maps, itinerary builder, and manages WebSocket connection state.
Backend Server	FastAPI + Async Python WebSockets RPC	Async event loop, WS route dispatcher, dual envelope RPC handler, ConnectionRegistry multi-client broadcaster.
AI Provider Layer	Gemini 1.5 Flash + Mock AI Fallback	Generates dynamic itineraries, day swaps, and weather replanning. 4s timeout instantly trips failover to Mock AI Provider.
Location Engine	Browser Geolocation + Overpass / OSM API	Fetches real-time POIs dynamically within 5000m GPS radius without hardcoded location biases.
Dual DB Store	PostgreSQL (Port 5432) SQLite (voyageai.db)	Primary PostgreSQL connection with automatic fallback to local SQLite file database. Thread-safe scoped sessions per frame.

2. Real-Time Data Flow & Failover Architecture

VoyageAI uses a resilient WebSocket RPC communication mechanism paired with automatic service failovers to guarantee 100% platform availability:



Core Resiliency & Protocol Features:

- **Dual Envelope WS Support:** Compatible with {id, action, payload} and legacy {type, reqname, data, requestId} formats.
- **Client Reconnection Engine:** Implements exponential backoff with jitter (1s to 30s max, 1.5 multiplier) and an offline request buffer (offlineBuffer) that auto-flushes upon socket reconnection.
- **AI Fault Tolerance:** Gemini 1.5 HTTP requests carry a strict 4.0-second timeout. If Gemini is rate-limited (HTTP 429) or times out, the backend seamlessly returns deterministic Mock AI data without breaking client contracts.
- **Database Failover:** Attempts PostgreSQL local socket connection first; automatically falls back to SQLite file storage if PostgreSQL service is offline.

3. Technical Skills Used in Project

This project integrates skills across modern full-stack development, AI system engineering, real-time networking, geospatial computing, and database design:

Category	Skills & Technologies	Application Context in VoyageAI
Frontend Engineering	React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons, Context API	Built single-page React app with TypeScript type safety, custom CSS design system (#1F2522, #355F58), reactive trip state management.
Real-Time WebSockets	WebSocket RPC, Exponential Backoff, Offline Buffering, ConnectionRegistry	Engineered resilient bi-directional socket client/server RPC with auto-reconnect, jitter, offline replay queue, and push event broadcasting.
AI Engineering & Prompts	Google Gemini 1.5 Flash, Prompt Engineering, Persona Isolation, Fallbacks	Designed location-agnostic production prompts, persona guardrails (Local Tour Guide vs Trip Operator), structured JSON schema outputs.

Geospatial & Mapping	Browser Geolocation API, OpenStreetMap / Overpass API, Leaflet Maps	Implemented live physical GPS coordinate detection with 5000m radius bounding-box queries for dynamic real-time local place discovery.
Backend & Data Architecture	Python 3.10+, FastAPI, Asyncio, SQLAlchemy, Dual DB (PostgreSQL + SQLite)	Developed async backend server with dual database auto-failover, thread-safe session scoping per WS message frame, and REST endpoints.
DevOps & System Quality	Environment Isolation, Virtualenv, Pydantic, HTTP Status Handlers	Established strict error handling, schema validation, zero-warning TypeScript builds, and isolated virtual environments.

Summary Conclusion: *VoyageAI represents a modern, resilient full-stack architecture built to deliver fast, reliable, location-aware travel planning with high fault tolerance and scalable real-time capabilities.*